

Climate Data Signal Processing

Rain Zhao

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Data used for this project were gathered from the Canadian Hydrographic Service, Methane concentration levels over time were extracted for time seise analysis.

Part a:

The poles and zeros can be represented as follows:

$$\text{Poles} : q = e^{\frac{i2\pi f_0}{f_s}} ; \text{Zeros} : p = (1 + \epsilon)q \quad (1)$$

Where poles are located on the unit circle, zeros are outside of the unit circle.

This system is unstable because the poles are outside of the circle when $\epsilon > 0$, the filter is real since the poles and zeros occur in conjugate pairs of each other.

Part b:

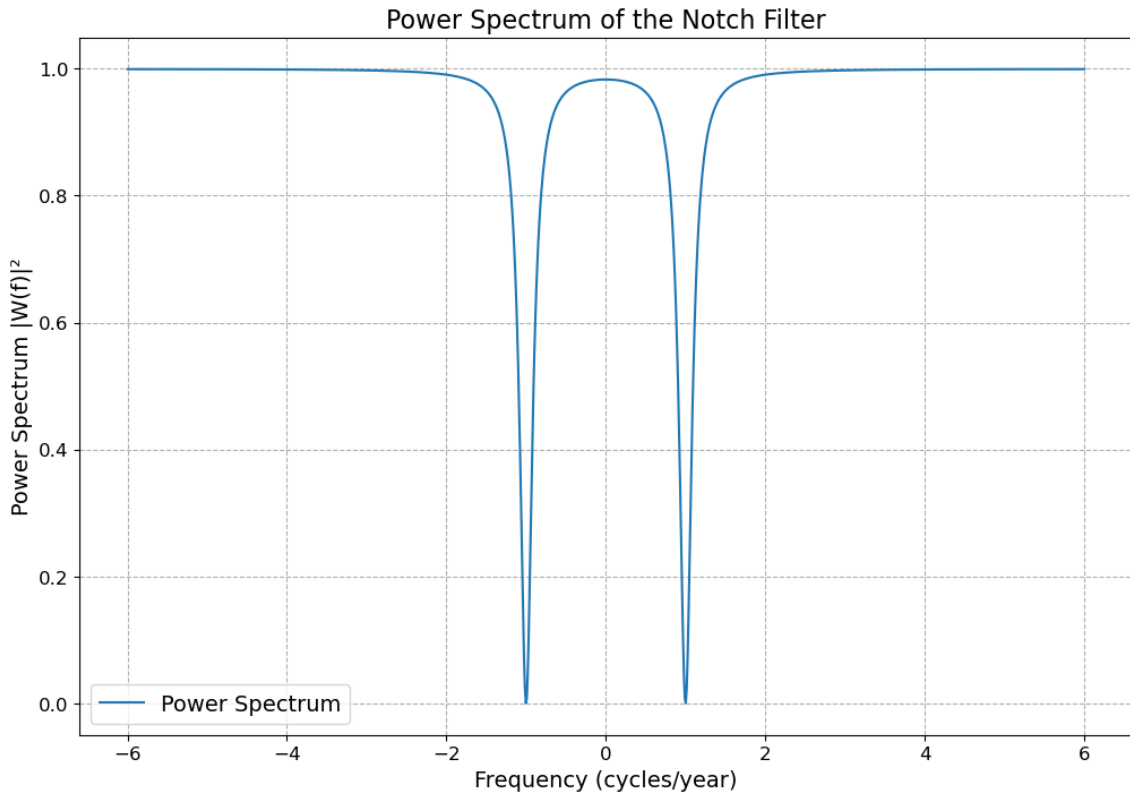


Figure 1: The power spectrum of $|W(f)|^2 = W(f)W(f)^*$.

Part c:

Based on the plot, the half level would be $1/2=0.5$, and if we were to draw a dashed line, we would expect a separation of 0.2 between the intersections, therefore the FWHM would be 0.2 [cycles/year]. To make the notches sharper, we should make the value of epsilon smaller, since then the poles would move closer to the unit circle.

Part d:

For $N(z)$, we have the numerator terms from $W(z)$ expanded as follows:

$$M(z^{-1} - q)(z^{-1} - q^*) = M[z^{-2} - 2\cos(\frac{2\pi f_0}{f_s})z^{-1} + 1] \quad (2)$$

With:

$$q + q^* = 2\cos(\frac{2\pi f_0}{f_s}); qq^* = 1 \quad (3)$$

Thus, we have $a = M$, $b = -2M\cos(\frac{2\pi f_0}{f_s})$, $c = M$

Likewise, the denominator $D(z)$ can be expanded in to the following with $p = (1 + \epsilon)q$:

$$(z^{-1} - p)(z^{-1} - p^*) = (z^{-1} - (1 + \epsilon)q)(z^{-1} - (1 + \epsilon)q^*) \quad (4)$$

Comes down to:

$$z^{-2} - 2(1 + \epsilon)\cos(\frac{2\pi f_0}{f_s})z^{-1} + (1 + \epsilon)^2 \quad (5)$$

Which when normalized gives: $F = -2(1 + \epsilon)\cos(\frac{2\pi f_0}{f_s})$, $G = (1 + \epsilon)^2$

Part e:

See python code file for function $ratFilter(N, D, x)$.

Part f:

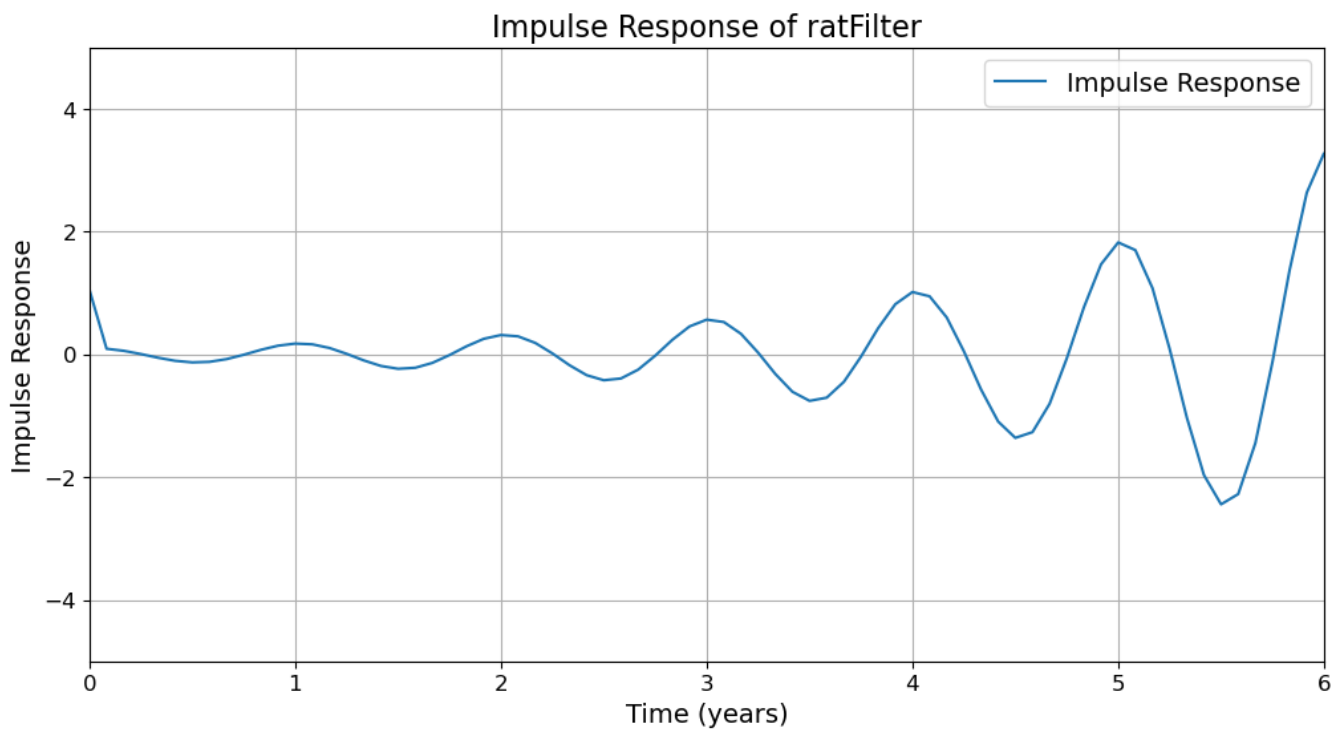


Figure 2: The plotted impulse response from 0 to 6 years.

Part g:

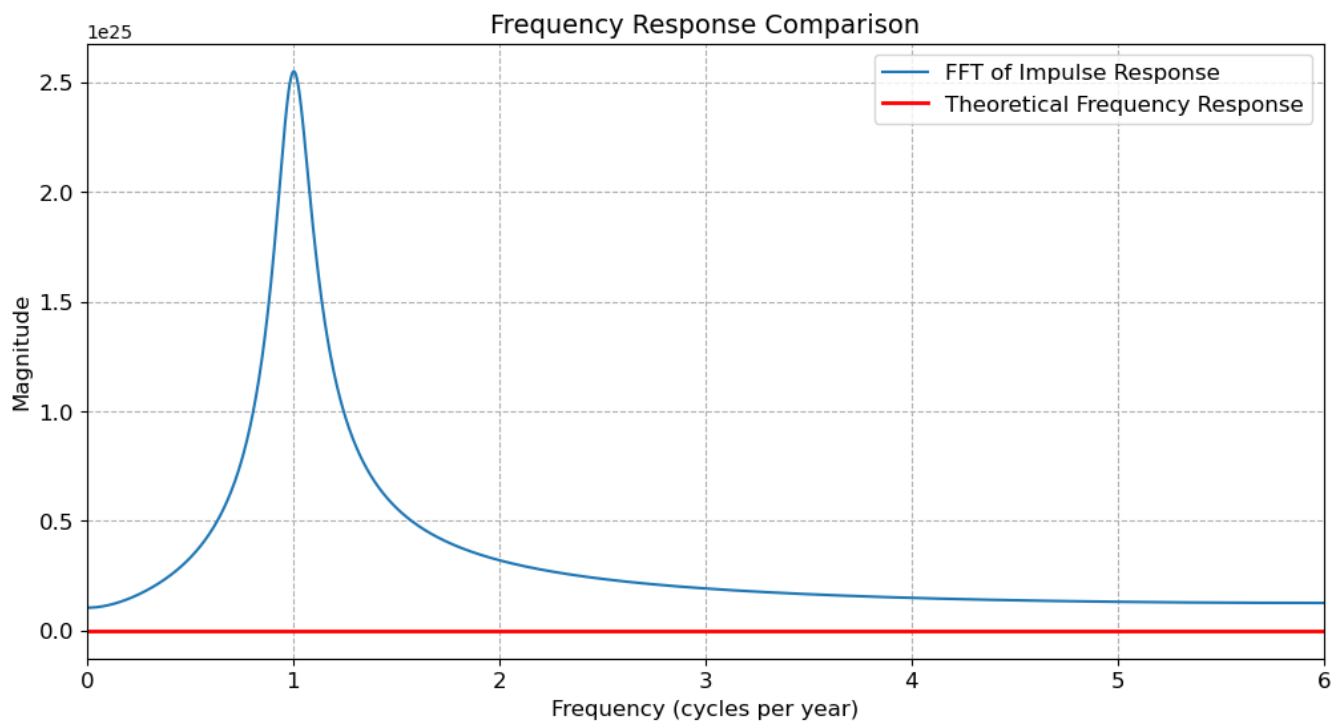


Figure 3: The Fourier transformed impulse response, plotted from 0 to 6 cycles per year.

Question 2:

Collaborators: None

Use of AI: None

Part a:

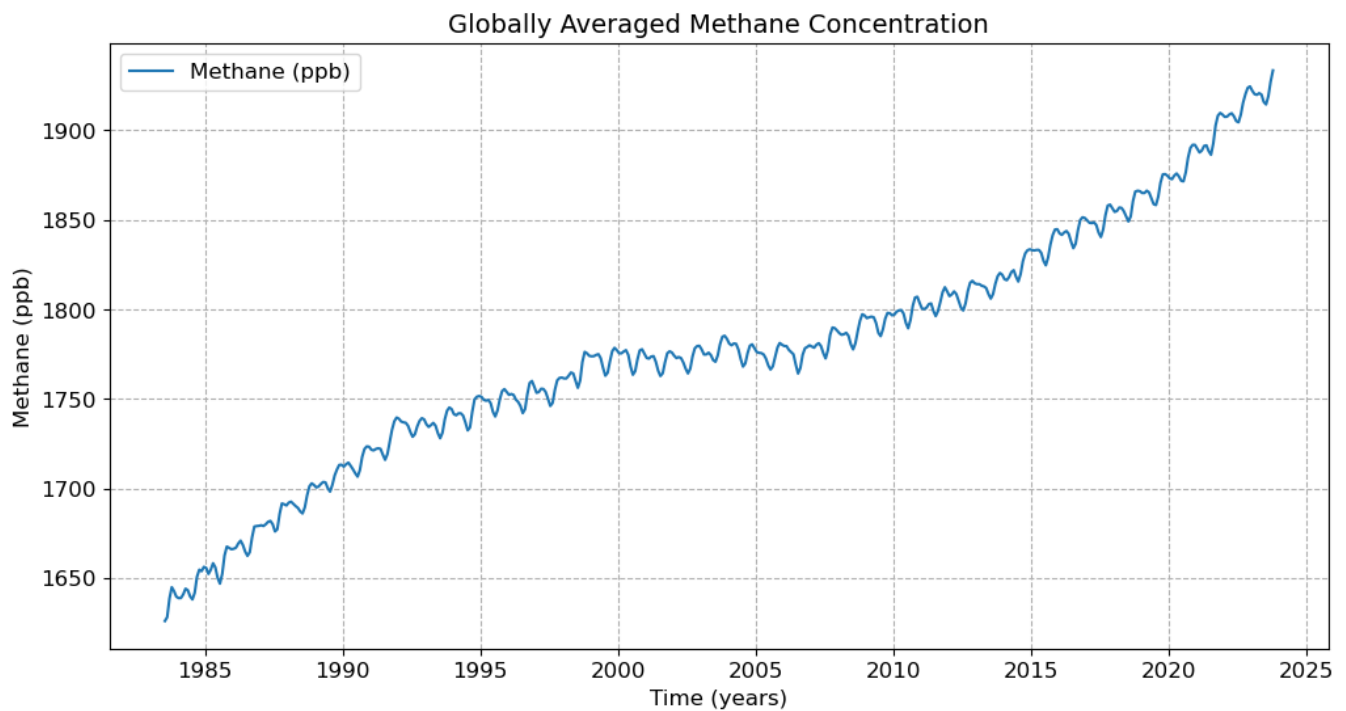


Figure 4: The original data of the globally averaged methane values.

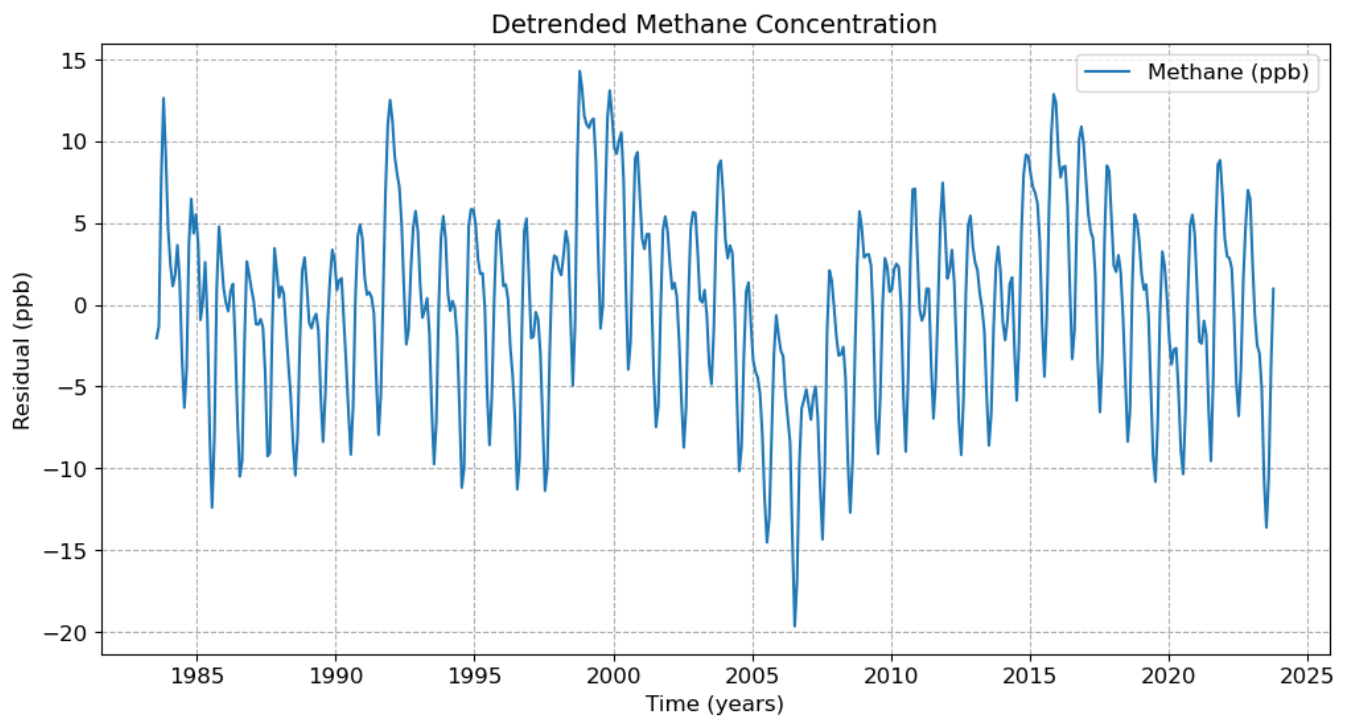


Figure 5: The de-trended data of the globally averaged methane values, before applying filters.

Part b:

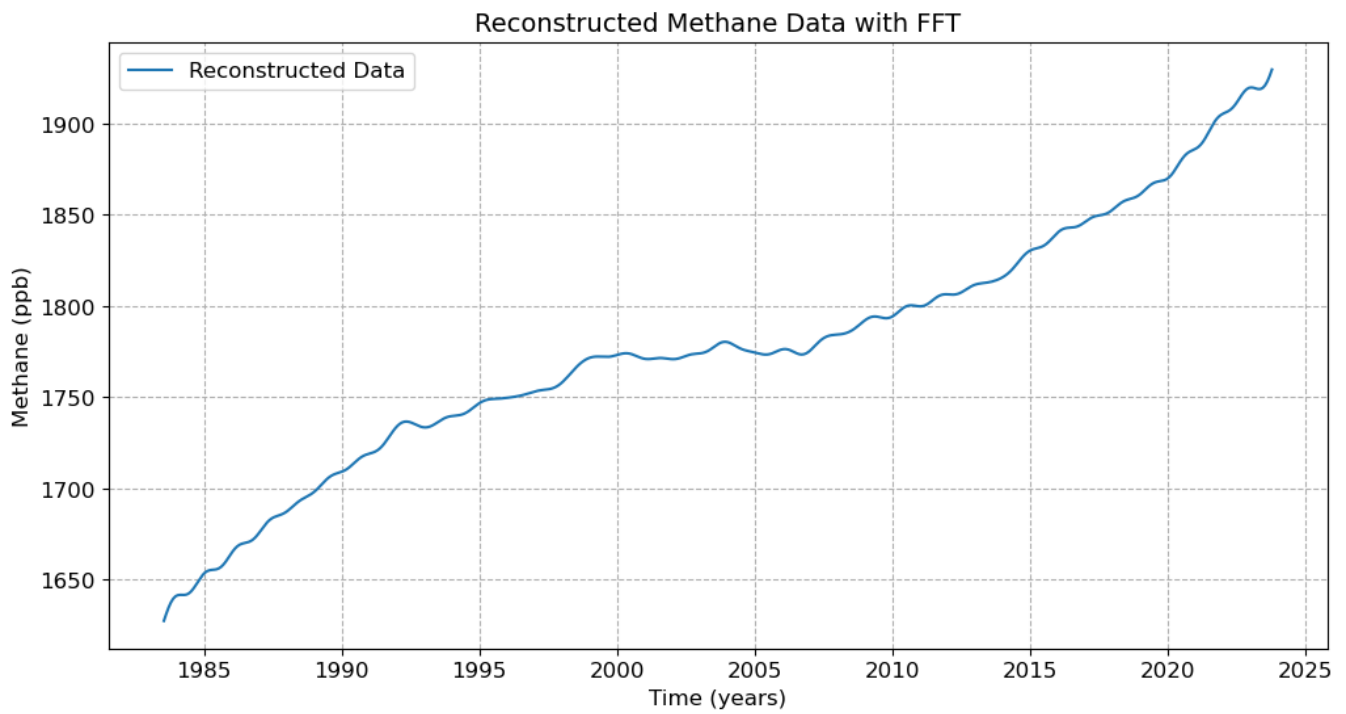


Figure 6: The reconstructed data after filtering with annual oscillations removed.

Part c:

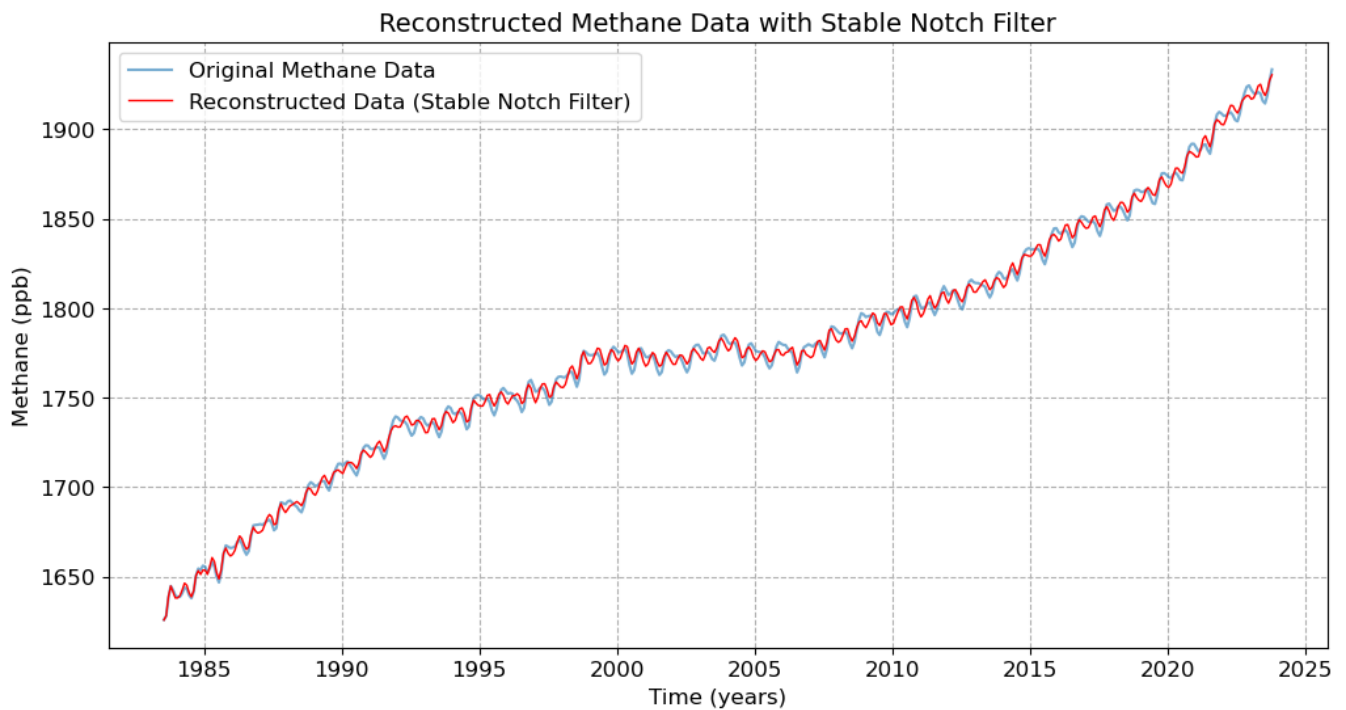


Figure 7: The reconstructed data after applying the designed notch filter, plotted over the original data.

Part d:

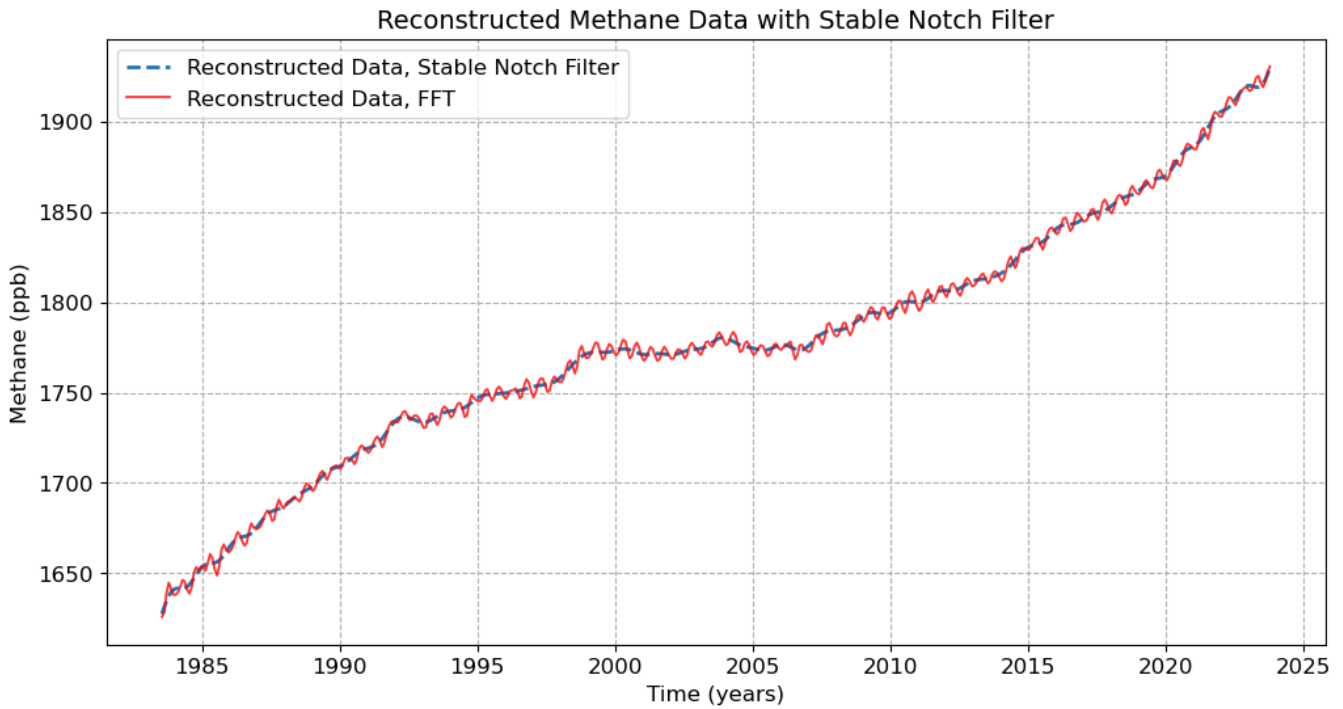


Figure 8: **Both FFT, and notch filtered datasets plotted on the same plot, with different lifestyle and colors.**

Fourier filtering transforms the entire detrended time series to the frequency domain, zeroing out the spectral components corresponding to frequencies above the annual cycle, then inverse-transform to filter. Some advantages would be the ability to explicitly remove or attenuate frequency bands, and the great performance with evenly spaced data. Disadvantages: Fourier filtering cannot be used for real-time processing, and may introduce edge effects at the boundaries due to FFT, it also doesn't work well when the signal is non-stationary, having changes gaps can lead to spectral leakages like we looked at before. Designing and applying a notch filter in the time domain removes the targeted annual frequency from the detrended signal, then gives us the ability to add back the multi-annual trend in reconstructing the full signal. Some advantages of this would be the ability to operate on the time-domain data, which is more suitable for real-time processing. It also has more flexibility in causal form and can be adapted for non-stationary signals (with appropriate design). Most importantly, it can precisely target unwanted frequencies. Disadvantages: The filter design takes more effort and time compared to the Fourier filtering, a stable notch filter is important in reconstructing the data. This method also depends sensitively on parameters, such as epsilon and normalization factor, making tuning critical throughout the application. The operation in time domain could also introduce phase shifts and distortions. To summarize, The Fourier method may have a smoother trend removal but sometimes obscure transient features, where the notch filter method gives more localized removal at the cost of potential design challenges.

Part e:

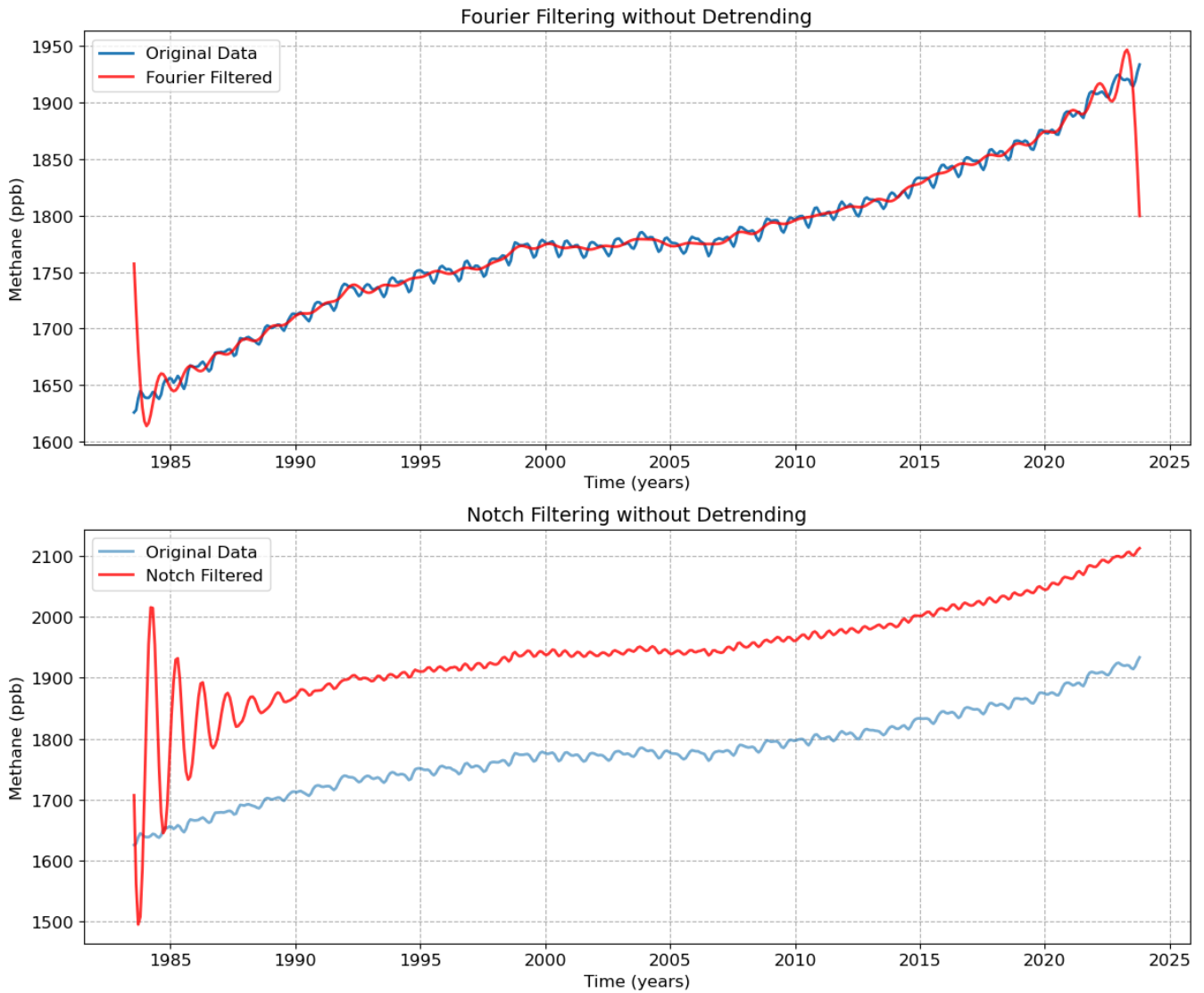


Figure 9: Both of the filtering repeated without de trending the data, we can see significant tailing on both ends of the FFT filtering plot, and a massive oscillation at the beginning of the Notch filtering plot, these are both caused by the absence of de-trending.

The de-trending has effects of: Isolating the oscillations: removing the long-term, low-frequency component(trend), isolating the higher frequency oscillations, leaving room for better filtering operations for both methods. Improving accuracy greatly. Reconstruction: when reconstructing without de-trending, the distortions will keep existing, thus misrepresenting the true behavior. However, de-trending takes an assumed pattern and shape, which sometimes do not fully represent the data. For FFT filtering, no de-trending causes none clear removal of seasonal oscillations. On the other hand, the notch filter ended up overreacting to the raw data without de-trending, creating distortions.

Question 3:

Collaborators: None

Use of AI: None

Part a:

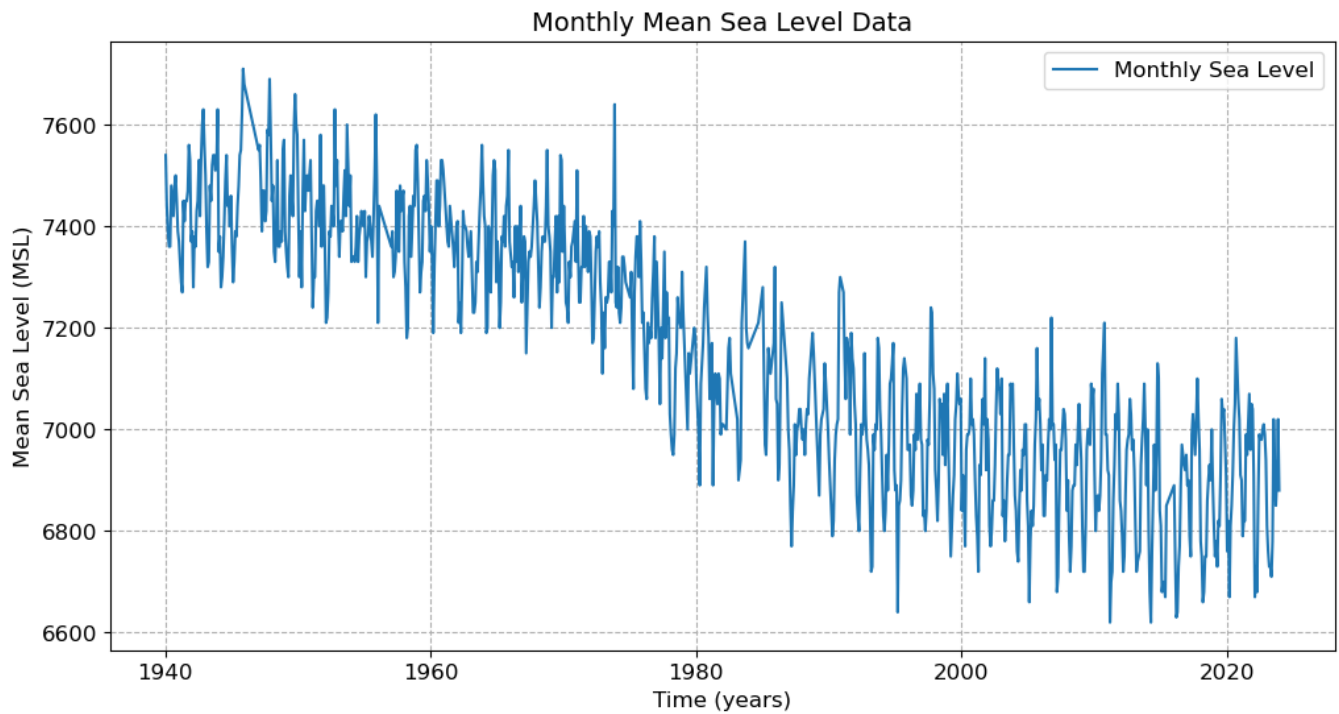


Figure 10: The loaded and plotted raw data of the Tide Gauge Data at CHURCHILL station, with only a few non-contiguous missing months of data. (<https://psmsl.org/data/obtaining/rlr.monthly.data/447.rlrdata>)

Part b:

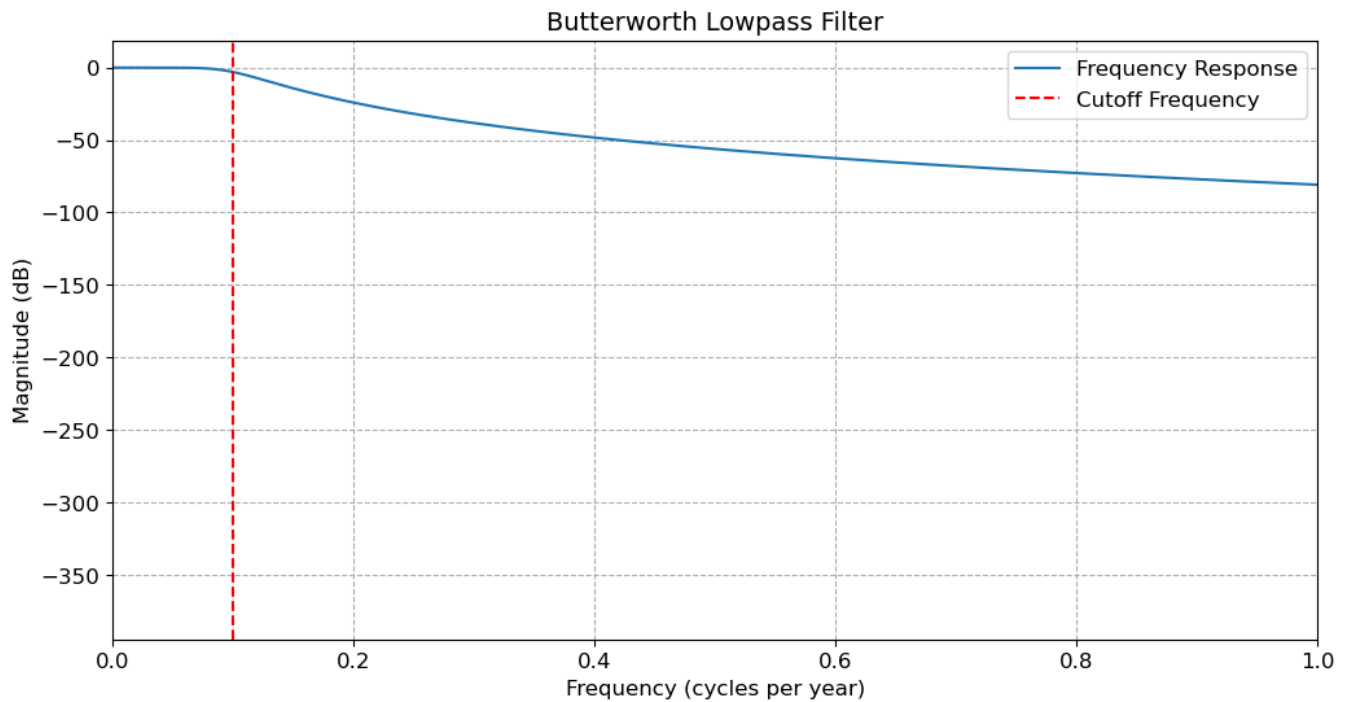


Figure 11: The frequency response of my designed filter to create a low-pass filter, in deciBels against frequency.

I chose a Butterworth filter because it gives a maximally flat response in the pass-band, meaning that oscillations with periods longer than 10 years are preserved with minimal distortion. I picked the 4th order to balance a smooth roll-off. This isolates the decadal variability in the mean sea level data.

Part c:

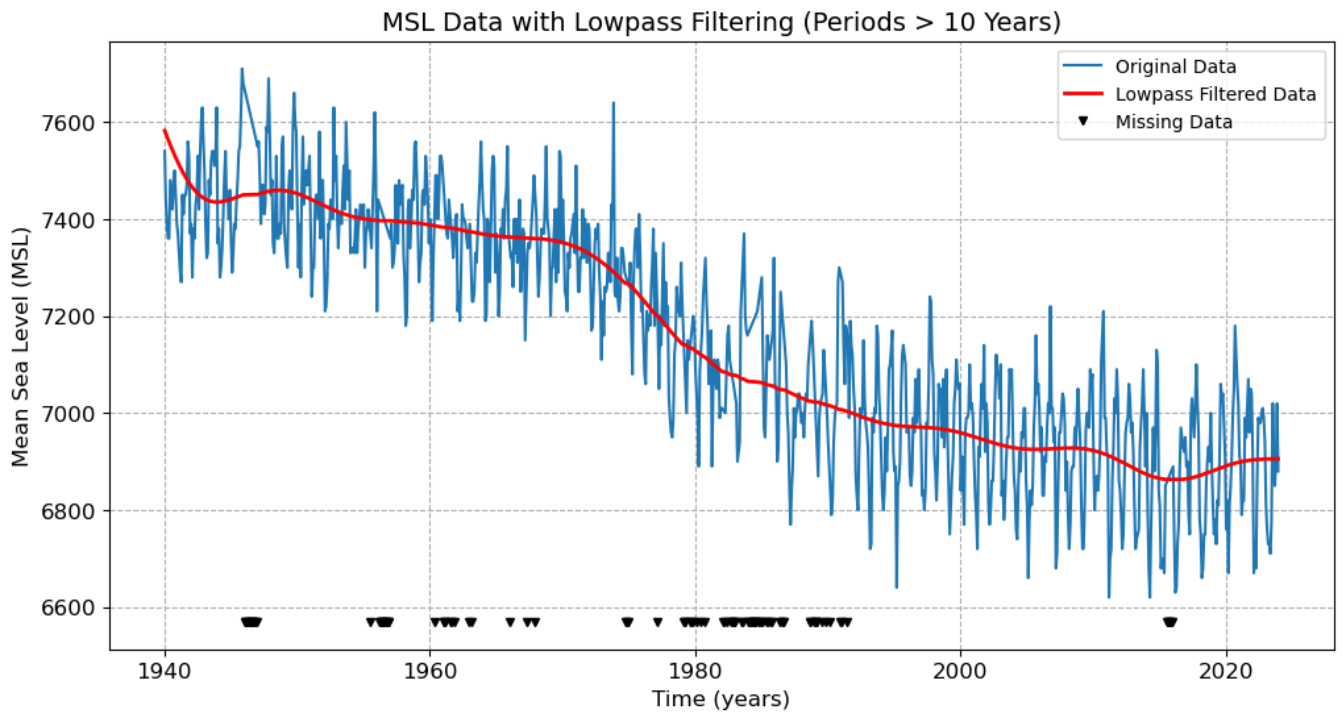


Figure 12: The plotted low-pass filtered dataset(red) over the original dataset, with markers indicating the missing values in time.

Part d:

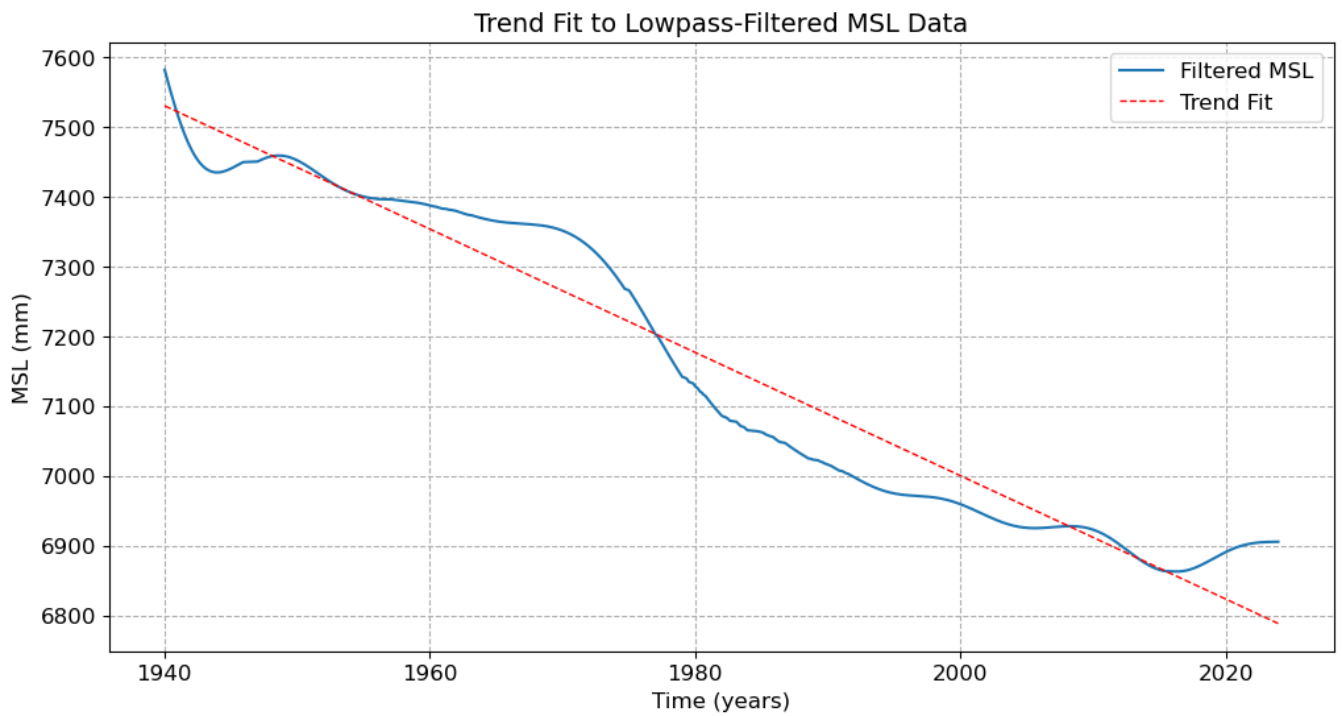


Figure 13: The low-pass filtering fitted with a 1 degree polynomial, finding a approximate rate of $-8.85[mm/yr]$ of sea level change, over a time range around 84 years.

References

C. Lee (2025) PHY 408 Filters; *Documentation*. Physics Department; University of Toronto.

Canadian Hydrographic Service (2025) CHURCHILL Tide Gauge Data (447); *Dataset*. Permanent Service for Mean Sea Level; National Oceanography Center.
<https://psmsl.org/data/obtaining/stations/447.php>